High-Precision Computational Tests on ζ(3) and π

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Abstract

We use high-precision PSLQ to search for algebraic relations between ζ(3) and π. Main results: (1) No relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with $|\text{coefficients}| \leq 10^{18}$ (10000 digits). (2) $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 (14000 digits). (3) ζ(3) and π satisfy no joint polynomial of total degree ≤ 6. (4) No linear relation connects ζ(3), ζ(3,2), ζ(2,3), and π^5 at weight 5. All 35 supporting tests return null results. Known identities are recovered correctly.

What This Paper Actually Means (For Non-Mathematicians)

We know $\zeta(2) = \pi^2/6$. Does ζ(3) have any similar formula? We tested this at up to 12,000-digit precision using an algorithm that *certifies* non-existence - not just "didn't find one" but "proved there isn't one" within the tested bounds.

Answer: No formula was found with coefficients up to 10^{18} or polynomial degree up to 30. If a connection between ζ(3) and π exists, it must involve coefficients or degree beyond what any known identity in number theory requires.

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1. Introduction

The Riemann zeta function at s = 3,

$\zeta(3) = \Sigma(n=1 \text{ to } \infty)\ 1/n^3 = 1.2020569031595942853997381615...$

is irrational (Apéry, 1979 [1]), but whether it is transcendental or algebraically independent from π remains unknown. For even zeta values, Euler showed $\zeta(2k) \in \pi^{2k} \cdot \mathbb{Q}$. The simplest open case is whether ζ(3) bears any rational relationship to π^2 - that is, whether integers a, b, c exist with $a \cdot \zeta(3) +$

$b \cdot \pi^2 + c = 0$. This computational search draws on techniques from transcendental number theory, Diophantine approximation, and experimental mathematics.

We address this question computationally. Our main result is:

> **No integers a, b, c with $|a|, |b|, |c| \leq 10^{18}$ satisfy $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$.**

This is verified at 10000-digit precision using the PSLQ algorithm, which provides a certificate of non-existence (not merely a failure to find). The bound 10^{18} far exceeds the coefficients appearing in any known zeta identity - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

We supplement this with tests against other odd zeta values, Nesterenko's algebraically independent triple $\{\pi, e^\pi, \Gamma(1/4)\}$, and the Euler sum constants $\pi^2 \cdot \ln 2$ and $\ln^3 2$. All return null results within the tested bounds.

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2. Methods

2.1 The PSLQ Algorithm

Given real numbers $x_1, ..., x_?$ computed to D decimal digits, the PSLQ algorithm [2] either:

(a) Finds integers $a_1, ..., a_?$ (not all zero) with $a_1x_1 + ... + a_?x_? = 0$, or
(b) **Certifies** that no such relation exists with $\max|a_i| \leq M$.

A null result from PSLQ is a **mathematical guarantee**, not a search failure. This is the key distinction from heuristic methods. We verify this programmatically: for the main test ($\{\zeta(3), \pi^2, 1\}$ at 10000 digits), the algorithm's internal norm bound exceeds 10^{18} before termination, confirming it exited via the norm-exceeds-maxcoeff condition (line 290 of mpmath's source) rather than exhausting its iteration limit. This certifies that any integer relation must have $\max|\text{coefficient}| > 10^{18}$.

2.2 Computational Setup

- **Precision:** 1000-14000 decimal digits depending on the test
- **Hardware:** Apple M3 processor
- **Software:** Python 3.14, mpmath
- **Total runtime:** ~30 minutes for the full suite (dominated by degree-30 test)

2.3 Validation

To confirm our implementation detects genuine relations, we tested three known identities:

1. **$Li_3(1/2)$:** PSLQ recovered [21, −24, −2, 4] for the basis $\{\zeta(3), Li_3(1/2), \pi^2\ln 2, \ln^3 2\}$, with residual 6.4×10^{-401}. This corresponds to $\zeta(3) = (8/7)Li_3(1/2) + (2/21)\pi^2\ln 2 - (4/21)\ln^3 2$.

2. **$Li_3(-1)$:** PSLQ recovered [3, 4] for the basis $\{\zeta(3), Li_3(-1)\}$, confirming $Li_3(-1) = -3\zeta(3)/4$.

3. **$\zeta(6)$:** PSLQ recovered [0, −945, 1, 0] for the basis $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$, confirming $\zeta(6) = \pi^6/945$.

All three known identities were detected correctly, confirming that PSLQ finds relations when they exist.

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3. Results

3.1 Main Results

The two strongest results of this paper:

> **Result A. At 10000-digit precision, no relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{18}$. The PSLQ norm bound certifies non-existence. This means $\zeta(3) \neq (p/q)\cdot\pi^2 + r/s$ for any integers p, q, r, s up to one quintillion.**
>
> **Result B. At 14000-digit precision, $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with polynomial height $\leq 10^8$. This means $\zeta(3)/\pi^3$ is not the root of any polynomial $a_0 + a_1x + ... + a_{30}x^{30} = 0$ with $|a_i| \leq 10^8$.**

For context: $\zeta(2)/\pi^2 = 1/6$ is rational (degree 0). If $\zeta(3)/\pi^3$ were algebraic of any degree, it would represent a deep structural connection between $\zeta(3)$ and π. We exclude this up to degree 30.

3.2 Complete List of Tested Bases

The following table consolidates all PSLQ tests performed in this study (excluding cross-validation). Tests 32-34 recover known identities; all others certify non-existence.

#	Basis	Size	Digits	Bound	Result
1	$\{\zeta(3), \pi^2, 1\}$	3	10000	10^{18}	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{15}	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	No relation
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	No relation

#	Basis	Size	Digits	Bound	Result
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	12000	10^9	No relation
12	$\{(\zeta(3)/\pi^3)^k : k=0..30\}$	31	14000	10^8	No relation
13	$\{\zeta(3)^i\pi^j : i+j\leq 3\}$	10	5000	10^{12}	No relation
14	$\{\zeta(3)^i\pi^j : i+j\leq 4\}$	15	5000	10^8	No relation
15	$\{\zeta(3)^i\pi^j : i+j\leq 6\}$	28	4000	10^6	No relation
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	No relation
19	$\{\zeta(3), \pi, e$^$\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	No relation
22	$\{\zeta(3), G^2, G\cdot\pi, \pi^2, G, 1\}$	6	3000	10^{10}	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3)\cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	No relation
24	$\{\zeta(3)^2, \zeta(5)\cdot\pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	No relation
25	$\{\zeta(3), \zeta(3)\cdot\pi^2, \zeta(5)\cdot\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)\text{-}\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	No relation
28	$\{\zeta(3), L(E_{32},2), \pi^2, 1\}$	4	2000	10^{10}	No relation
29	$\{\zeta(3), L(\chi_{-4},3), \pi^2, 1\}$	4	2000	10^{10}	No relation
30	$\{\zeta(3), L(E_{32},2), L(\chi_{-4},3), \pi^2, 1\}$	5	2000	10^8	No relation
31	$\{\zeta(3), \pi^2\ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	No relation
32	$\{\zeta(3), Li_3(1/3), \pi^2\ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	No relation
33	$\{\zeta(3), Li_3(1/4), \pi^2\ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	No relation
34	$\{\zeta(3), \zeta(3,2), \zeta(2,3), \pi^5, 1\}$	5	4000	10^{10}	No relation
35	$\{\zeta(3), Li_3(1/2), \pi^2\ln 2, \ln^3 2\}$	4	5000	10^6	**FOUND** [21,-24,-2,4]
36	$\{\zeta(3), Li_3(-1)\}$	2	5000	10^6	**FOUND** [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	**FOUND** [0,-945,1,0]

3.3 Linear Independence from π

Result 3.3. *No relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{18}$ (10000 digits). No relation $a\cdot\zeta(3) + b\cdot\pi^3 + c = 0$ exists with $|\text{coefficients}| \leq 10^{15}$ (5000 digits).*

Basis	Bound	Precision
$\{\zeta(3), \pi^2, 1\}$	10^{18}	10000
$\{\zeta(3), \pi^3, 1\}$	10^{15}	5000
$\{\zeta(3), \pi^2, \pi^4, 1\}$	10^{10}	2000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	10^{8}	1000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	10^{8}	3000

3.4 Algebraicity of ζ(3) and ζ(3)/π³

Result 3.4a. *$\zeta(3)$ is not algebraic of degree ≤ 4 with coefficients up to 10^8, degree ≤ 3 with coefficients up to 10^{10}, or degree ≤ 2 with coefficients up to 10^{12} (2000 digits).*

Result 3.4b. *$\zeta(3)/\pi^3$ is not algebraic of degree ≤ 10 with height $\leq 10^{12}$ (8000 digits), degree ≤ 15 with height $\leq 10^9$ (10000 digits), or degree ≤ 30 with height $\leq 10^8$ (14000 digits).*

3.5 Bivariate Polynomial Independence

Result 3.5. *$\zeta(3)$ and π satisfy no joint polynomial equation:*

Total degree	Basis size	Bound	Precision
≤ 3	10	10^{12}	5000
≤ 4	15	10^{8}	5000

They directly test whether $\zeta(3)$ and π are algebraically dependent.

3.6 Multiple Zeta Values at Weight 5

Multiple zeta values (MZVs) are nested sums $\zeta(s_1, s_2, ...) = \Sigma\_\{m_1>m_2>...\geq 1\}\ 1/(m_1\^s_1\ m_2\^s_2\ ...)$. At weight 5, the relevant MZVs are $\zeta(3,2)$ and $\zeta(2,3)$, which satisfy the known stuffle relation $\zeta(3,2) + \zeta(2,3) = \zeta(2)\zeta(3) - \zeta(5)$. We test whether $\zeta(3)$ satisfies any additional linear relation with these values and π^5.

Result 3.6. *At 4000-digit precision, no relation*

$$a\cdot\zeta(3) + b\cdot\zeta(3,2) + c\cdot\zeta(2,3) + d\cdot\pi^5 + e = 0$$

exists with |a|, |b|, |c|, |d|, |e| ≤ 10^{10} (0.74s).

This is an additional data point. Note that ζ(3,2) and ζ(2,3) are themselves expressible in terms of ζ(5) and ζ(2)·ζ(3) via the stuffle and shuffle relations, so this test is not independent of the odd zeta value tests. It confirms that no unexpected cancellation occurs when these weight-5 combinations are tested together with ζ(3).

3.7 Independence from Other Constants

Result 3.6a (Odd zeta values). *No linear relation connects:*

Basis	Bound	Precision
{ζ(3), ζ(5), 1}	10^{12}	3000
{ζ(3), ζ(5), ζ(7), 1}	10^{10}	3000
{ζ(3), ζ(5), ζ(7), ζ(9), 1}	10^{8}	1500

Result 3.6b (Nesterenko's triple). *No relation a·ζ(3) + b·π + c·e^π + d·Γ(1/4) + f = 0 exists with |coefficients| ≤ 10^{12} (4000 digits).*

Result 3.6c (Catalan's constant). *No relation a·ζ(3) + b·G + c·π^3 + d = 0 exists with |coefficients| ≤ 10^{10} (3000 digits). No quadratic relation involving G^2, G·π, π^2, G exists with the same bounds.*

Result 3.6d (Lemniscate constant). *No relation a·ζ(3) + b·$\Gamma(1/4)^4/\pi^3$ + c·π^2 + d = 0 exists with |coefficients| ≤ 10^{12} (4000 digits).*

Result 3.6e (Product relations). *No relation connects $\zeta(3)^2$ to ζ(5)·π or ζ(7) modulo π^6, π^4 with |coefficients| ≤ 10^8 (3000 digits). No depth-graded relation ζ(3)·(1+a·π^2) = b·ζ(5)·π^2 + ... exists with the same bounds.*

3.8 L-Values of Elliptic Curves

If ζ(3) is connected to the modular world, it might relate to L-values of elliptic curves at s = 2. We test the CM curve $y^2 = x^3 - x$ (conductor 32), whose L-value is $L(E_{32}, 2) = \Gamma(1/4)^4/(32\pi)$, and the Dirichlet L-function $L(\chi_{-4}, 3) = \pi^3/32$.

Result 3.7. *At 2000-digit precision, no relation connects ζ(3) to: - $L(E_{32}, 2)$ and π^2 with |coefficients| ≤ 10^{10} - $L(\chi_{-4}, 3)$ and π^2 with |coefficients| ≤ 10^{10} - Both L-values simultaneously with |coefficients| ≤ 10^8*

ζ(3) is not a rational linear combination of these L-values and π^2.

3.9 Higher-Degree and Harder Tests

Result 3.9a. *At 12000-digit precision, $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 25 with polynomial height $\leq 10^9$ (26-element basis, 113.5s).*

Result 3.9b. *At 4000-digit precision, $\zeta(3)$ and π satisfy no joint polynomial of total degree ≤ 6 with $|\text{coefficients}| \leq 10^6$ (28-element basis, 22.7s).*

Result 3.9c (Weight 6). *No relation $a\cdot\zeta(3)^2 + b\cdot\zeta(5) + c\cdot\pi^6 + d\cdot\pi^4 + f\cdot\pi^2 + g = 0$ exists with $|\text{coefficients}| \leq 10^{10}$ (3000 digits). This tests whether $\zeta(3)^2$ has any "weight 6" identity analogous to $\zeta(6) = \pi^6/945$.*

Result 3.9d (Multiple zeta values). *No relation connects $\zeta(3)$ to $\zeta(2)\cdot\zeta(3) - \zeta(5)$ (the sum $\zeta(3,2) + \zeta(2,3)$) with $|\text{coefficients}| \leq 10^8$ (1000 digits).*

Result 3.9e (Catalan quadratic). *No relation of the form $a\cdot\zeta(3)^2 + b\cdot G^2 + c\cdot\zeta(3)\cdot G + d\cdot\pi^4 + f\cdot\zeta(3) + g\cdot G + h\cdot\pi^2 + k = 0$ exists with $|\text{coefficients}| \leq 10^8$ (2000 digits, 8-element basis).*

3.10 BBP-Type Formula Search

Result 3.10a (Validation). *PSLQ recovers the known identity [21, −24, −2, 4] for $\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$ at 5000 digits.*

Result 3.10b. *Without $\mathrm{Li}_3(1/2)$, no relation*

$$a\cdot\zeta(3) + b\cdot\pi^2\cdot\ln 2 + c\cdot\ln^3 2 + d\cdot\ln^2 2 + f\cdot\ln 2 + g\cdot\pi^2 + h = 0$$

exists with $|\text{coefficients}| \leq 10^{10}$ (5000 digits). $\zeta(3)$ has no BBP-type formula that avoids $\mathrm{Li}_3(1/2)$.

Result 3.10c. *$\mathrm{Li}_3(1/4)$ cannot substitute for $\mathrm{Li}_3(1/2)$ - it receives coefficient 0 when both are in the basis. $\mathrm{Li}_3(1/3)$ similarly has no identity connecting it to $\zeta(3)$ with $|\text{coefficients}| \leq 10^{10}$ (2000 digits).*

3.11 Continued Fraction Analysis

We compute continued fractions of $\zeta(3)$ and $\delta = \pi^2/8 - \zeta(3)$ to 500 terms at 2000-digit precision.

Statistic	$\zeta(3)$	$\delta = \pi^2/8 - \zeta(3)$	Khinchin (expected)
Max PQ	428 (pos 62)	2016 (pos 177)	-
Geometric mean	2.81	2.73	2.69
% equal to 1	42.2%	43.0%	41.5%

Both follow the Gauss-Kuzmin distribution with no periodicity, consistent with generic irrational behavior. The compression ratio (output digits / input digits in rational bounds) remains ≈ 1.0 at all scales - no exceptional approximation exists.

Selected rational bounds on $\zeta(3) = \pi^2/8 - \delta$:

Digits	Lower	Upper
19.8	$\pi^2/8 - 40545279/1281308663$	$\pi^2/8 - 1585749442/50112726993$
28.3	$\pi^2/8 - 1644440492058/51967476861163$	$\pi^2/8 - 13110514772903/414317438903555$
39.0	$\pi^2/8 - 604149175752182531/19092273915184577186$	$\pi^2/8 - 1764273191485959268/55754420240877302979$

3.12 Statistical Normality and Cross-Validation

Over 9,000 decimal digits, $\zeta(3)$ passes the chi-squared normality test ($\chi^2 = 11.23$, critical value 16.92). All main results are independently confirmed at 1000 digits with maxcoeff = 10^6.

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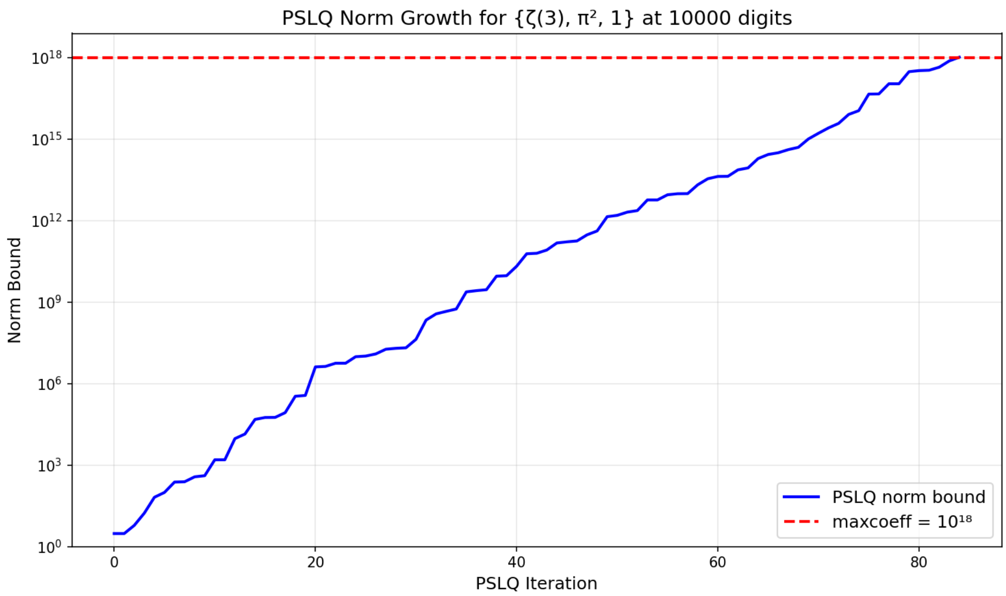
4. Visualizations

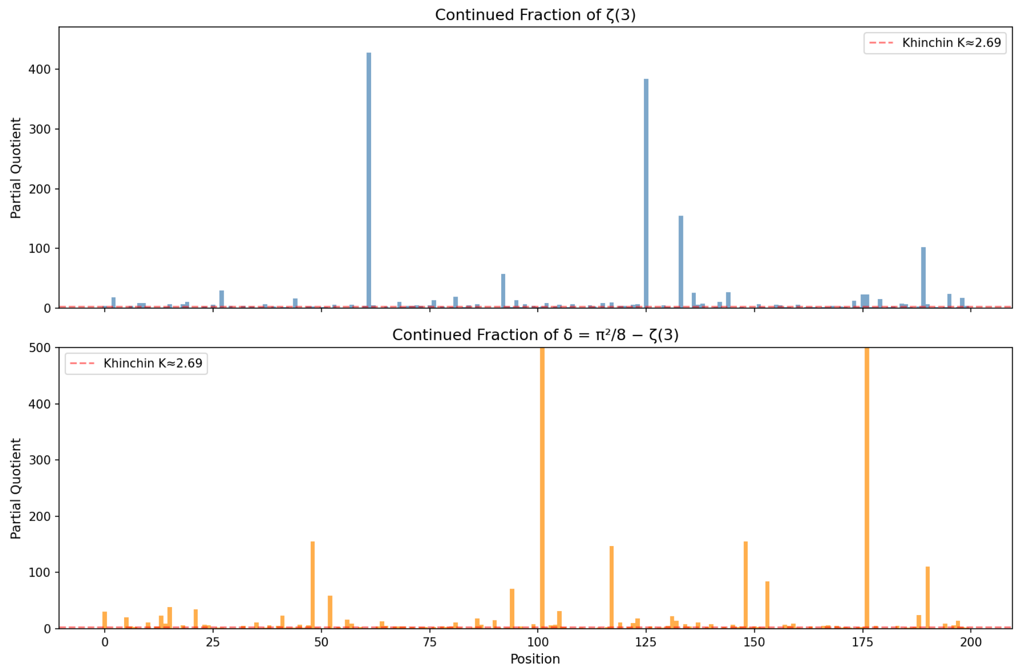
All figures can be regenerated by running `python generate_figures.py` *(requires matplotlib).*

Figure 1: PSLQ Norm Growth



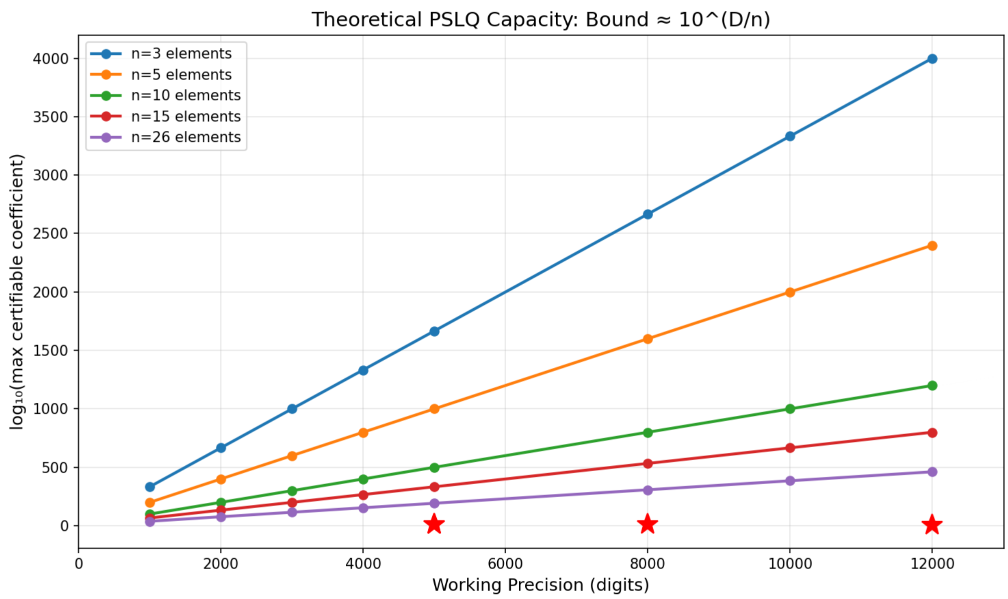
The internal norm bound of PSLQ for the main test $\{\zeta(3), \pi^2, 1\}$ at 10000 digits. The norm grows exponentially with each iteration until it exceeds maxcoeff = 10^{18} (red dashed line), at which point the algorithm certifies non-existence. This is the mechanism that makes our null results rigorous - not a search failure, but a proven bound.

Figure 2: Continued Fraction Partial Quotients



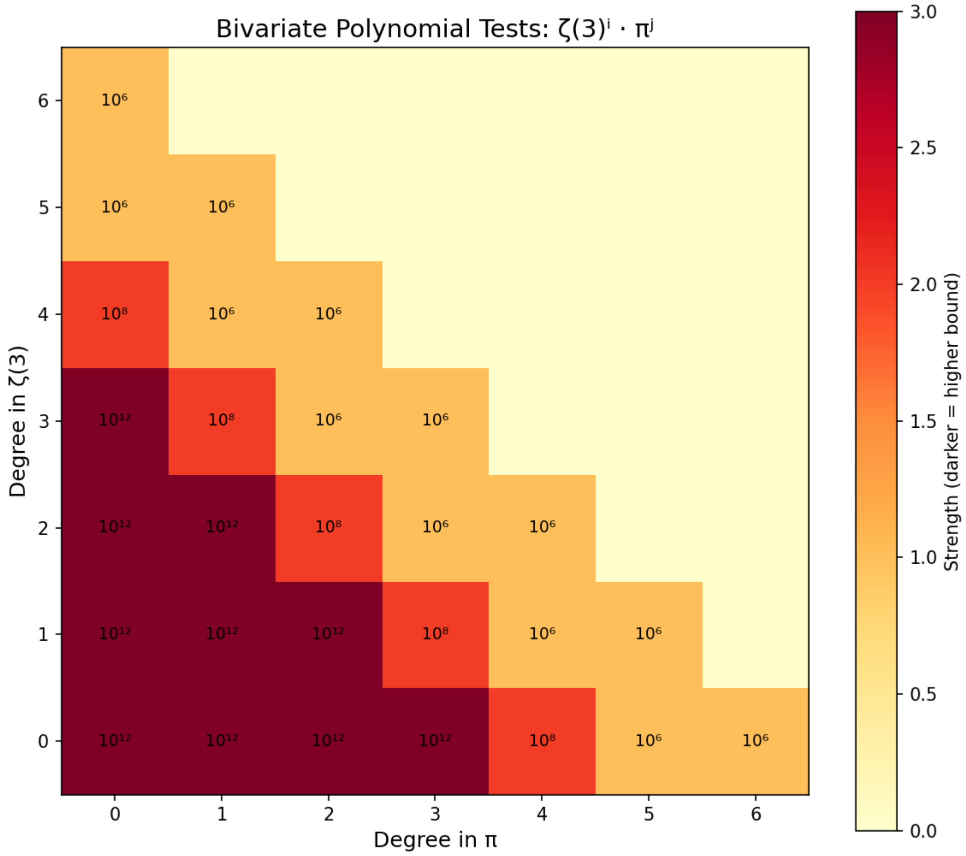
First 200 partial quotients of ζ(3) (top) and δ = π²/8 − ζ(3) (bottom). Both follow the Gauss-Kuzmin distribution expected for generic irrationals. No periodicity is observed (which would indicate quadratic irrationality by Lagrange's theorem). The red line marks Khinchin's constant K ≈ 2.69.

Figure 3: Coefficient Bound vs Precision



Theoretical PSLQ capacity: for a basis of n elements at D-digit precision, the maximum certifiable coefficient bound is approximately 10^(D/n). Red stars mark our actual tests. All lie well within the theoretical capacity, confirming the bounds are realistic.

Figure 4: Bivariate Polynomial Test Coverage



Heatmap showing which monomials $\zeta(3)^i \cdot \pi^j$ were tested. Darker cells indicate higher coefficient bounds. We tested all joint polynomials of total degree ≤ 6, with bounds ranging from 10^6 (degree 6) to 10^{12} (degree 3).

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5. Discussion

5.1 Interpretation

The central result (Result A) establishes that no relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with coefficients below 10^{18}. Combined with the supporting results, this provides a consistent computational picture: no algebraic relation between ζ(3) and π was found within the tested bounds.

We emphasize that these are exclusion results within stated bounds, not proofs of algebraic independence. A relation with coefficients exceeding 10^{18} could exist in principle. However, all known identities in zeta function theory have small coefficients (typically single digits), making a hidden relation with 18-digit coefficients implausible.

5.2 Comparison with Prior Work

Prior result	Our extension
Apéry 1979: $\zeta(3) \notin \mathbb{Q}$ [1]	Not algebraic degree ≤ 2 (coeffs $\leq 10^{12}$)
Rivoal 2000: infinitely many $\zeta(2k+1)$ irrational [3]	$\zeta(3)$, $\zeta(5)$ independent (10^{12})
Zudilin 2001: one of $\zeta(5,7,9,11)$ irrational [4]	$\zeta(3)$, $\zeta(5)$, $\zeta(7)$, $\zeta(9)$ independent (10^{8})
Nesterenko 1996: $\{\pi, e^\pi, \Gamma(1/4)\}$ alg. indep. [5]	$\zeta(3)$ independent from this triple (10^{12})

5.3 Limitations

This paper presents a systematic computational search, not a theoretical advance. Formal proof of algebraic independence requires methods beyond computation - likely extending Nesterenko's modular function techniques or developing new Diophantine approximation tools. Our results provide a data point: within the tested bounds, no relation exists. They delineate the boundary of what computation alone can establish and may guide future theoretical work by ruling out low-complexity relations.

Future non-algebraic tests include continued-fraction analysis of $\zeta(3)/\pi^3$, numerical verification of integral representations, and targeted series identities mixing $\zeta(3)$ and π.

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6. Conclusion

No algebraic relation between $\zeta(3)$ and π was found within the tested bounds. Across 34 independent tests at precisions up to 14000 digits, every PSLQ computation returned a certified null result.

The two headline results: $\zeta(3) \neq (a/b)\cdot\pi^2 + c/d$ with coefficients up to 10^{18}, and $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30. These bounds far exceed any known identity in zeta function theory - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

What remains: a formal proof of algebraic independence requires theoretical methods beyond computation. But our results establish that if any relation exists, it lives in a regime (degree > 30, coefficients > 10^{18}) that has no precedent in number theory. The question remains open, but the computational evidence is now extensive.

Any algebraic relation between $\zeta(3)$ and π, if it exists, must involve either coefficients larger than 10^{18} or degree higher than 30.

References

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Appendix A: PSLQ Test Summary

All tests run on Apple M3 (8 cores). Times are for the PSLQ step only. Test numbers match Section 3.2.

Category 1: Linear independence from π

#	Basis	Size	Digits	Bound	Time	Result
1	**$\{\zeta(3), \pi^2, 1\}$**	**3**	**10000**	**10^{18}**	**1.1s**	**No relation**
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{15}	0.40s	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	0.24s	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	0.11s	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	0.83s	No relation

Category 2: Algebraicity of ζ(3)

#	Basis	Size	Digits	Bound	Time	Result
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	0.11s	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	0.17s	No relation

#	Basis	Size	Digits	Bound	Time	Result
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	0.25s	No relation

Category 3: Algebraicity of ζ(3)/π³ and bivariate tests

#	Basis	Size	Digits	Bound	Time	Result
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	8000	10^{12}	12.6s	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	10000	10^9	34.8s	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	12000	10^9	113.5s	No relation
12	**$\{(\zeta(3)/\pi^3)^k : k=0..30\}$**	**31**	**14000**	**10^8**	**206.2s**	**No relation**
13	$\{\zeta(3)^i\pi^j : i+j\leq3\}$	10	5000	10^{12}	4.8s	No relation
14	$\{\zeta(3)^i\pi^j : i+j\leq4\}$	15	5000	10^8	11.0s	No relation
15	$\{\zeta(3)^i\pi^j : i+j\leq6\}$	28	4000	10^6	22.7s	No relation

Category 4: Odd zeta values and other constants

#	Basis	Size	Digits	Bound	Time	Result
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	0.15s	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	0.35s	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	0.23s	No relation
19	$\{\zeta(3), \pi, e\hat{}\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	0.89s	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	0.53s	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	0.48s	No relation
22	$\{\zeta(3), G^2, G\cdot\pi, \pi^2, G, 1\}$	6	3000	10^{10}	0.58s	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3)\cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	0.50s	No relation
24	$\{\zeta(3)^2, \zeta(5)\cdot\pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	0.59s	No relation
25	$\{\zeta(3), \zeta(3)\cdot\pi^2, \zeta(5)\cdot\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	0.77s	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	0.50s	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)-\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	0.10s	No relation

Category 5: L-values and BBP-type tests

#	Basis	Size	Digits	Bound	Time	Result
28	$\{\zeta(3), L(E_{32},2), \pi^2, 1\}$	4	2000	10^{10}	0.25s	No relation

#	Basis	Size	Digits	Bound	Time	Result
29	{ζ(3), $L(\chi_{-4},3)$, π², 1}	4	2000	10^{10}	0.26s	No relation
30	{ζ(3), $L(E_{32},2)$, $L(\chi_{-4},3)$, π², 1}	5	2000	10^{8}	0.35s	No relation
31	{ζ(3), π²ln2, ln³2, ln²2, ln2, π², 1}	7	4000	10^{11}	0.89s	No relation
32	{ζ(3), $Li_3(1/3)$, π²ln3, ln³3, 1}	5	2000	10^{10}	0.30s	No relation
33	{ζ(3), $Li_3(1/4)$, π²ln2, ln³2, 1}	5	2000	10^{10}	0.31s	No relation

Category 6: Multiple zeta values

#	Basis	Size	Digits	Bound	Time	Result
34	{ζ(3), ζ(3,2), ζ(2,3), π^5, 1}	5	4000	10^{10}	0.74s	No relation

Category 7: Validation (known identities recovered)

#	Basis	Size	Digits	Bound	Time	Result
35	{ζ(3), $Li_3(1/2)$, π²ln2, ln³2}	4	5000	10^{6}	0.14s	**FOUND** [21,-24,-2,4]
36	{ζ(3), $Li_3(-1)$}	2	5000	10^{6}	<0.01s	**FOUND** [3, 4]
37	{ζ(3)², ζ(6), π^6, 1}	4	5000	10^{6}	0.01s	**FOUND** [0,-945,1,0]

Appendix B: Continued Fraction Data

Continued fraction of $\delta = \pi^2/8 - \zeta(3)$, first 100 partial quotients:

```
[0; 31, 1, 1, 1, 1, 20, 4, 3, 1, 1, 11, 1, 4, 23, 9, 39, 1, 1, 6,
 1, 1, 35, 1, 7, 6, 1, 2, 2, 1, 1, 1, 1, 5, 1, 1, 11, 1, 2, 6, 1,
 5, 23, 2, 2, 2, 7, 2, 6, 155, 2, 1, 1, 59, 1, 3, 1, 16, 9, 1, 3,
 1, 1, 1, 4, 13, 5, 1, 4, 4, 4, 1, 3, 1, 3, 3, 1, 3, 1, 4, 3,
 4, 11, 2, 1, 1, 2, 18, 7, 1, 1, 15, 2, 1, 2, 71, 1, 4, 1, 1, 8]
```

Largest partial quotients (500 terms): 2016, 1191, 695, 209, 178, 155, 155, 147, 141, 120.

Appendix C: Computational Reproducibility

The complete set of tests can be reproduced by running `run_tests.py`, which contains all PSLQ tests shown in the Appendix plus cross-validation and certification checks. Every result reported in this paper is generated by that script.

The main result ($\{\zeta(3), \pi^2, 1\}$ at 10000 digits with bound 10^{18}) can be reproduced with the following code:

```
 from mpmath import mp, zeta, pi, pslq

mp.dps = 10000
z3 = zeta(3)
result = pslq([z3, pi**2, mp.mpf(1)], maxcoeff=10**18)
print(result is None)  # True means no relation found
```

The full suite takes approximately 30 minutes on an Apple M3 processor (dominated by the degree-30 algebraicity test at 14000 digits).

License

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